

VARDAAN BAJAJ

Machine Learning & Software Engineer

Computer vision · full-stack web · C/DSP systems | Jammu, India · open to remote

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SUMMARY

Machine learning and software engineer (B.Tech in Data Science and AI, 2026). First author of an IEEE CICT 2025 paper on detection and tracking in drone imagery. Built a browser-native BI platform and voice-feature Parkinson's research, and a C signal-processing simulation during a research internship at DRDO.

EXPERIENCE

Defence Research and Development Organisation (DRDO)

Hyderabad, India (Hybrid)

Research and Development Intern

Jan 2026 – Jun 2026

- Built a simulation framework in **C** for processing simulated time-series signals and telemetry data.
- Implemented an in-place double-precision **Radix-2 FFT** and spectral peak detection for chirp waveforms under noise.
- Modelled **RS-422** serial framing and a command-response state machine, using **C11 atomics** and multithreading for thread-safe state updates with minimal jitter.

AgryBin

Remote

Web/App Developer (Internship)

May 2025 – Aug 2025

- Handled all web and mobile development for an early-stage agritech startup as its sole developer, from requirements to delivery.
- Built the company's brand and product website.
- Built an Android app in **Flutter** with OTP login and modules for mandi prices, mandi requirements, news, transporter contacts and cold-storage contacts.

IIIT Naya Raipur × Mahyco

Naya Raipur, India

Research Collaboration (student team, faculty-supervised)

Aug 2024 – Dec 2024

- Built a results-review web app for a crop-imaging research collaboration between IIIT Naya Raipur and Mahyco.

PROJECTS

DataVista: Browser-Native BI Platform | React 18, TypeScript, Zustand, Dexie.js, TanStack Virtual

- Built an offline-first BI platform that runs in the browser: load CSV, TSV or XLSX files up to 100 MB and build pivots, calculated fields and dashboards without a backend.
- Built a recursive-descent **formula parser** for Excel-style calculated fields and window functions (ROW_NUMBER, RANK, DENSE_RANK, rolling averages), with Vitest and Playwright tests in GitHub Actions.

NeuroInsight-AI: Voice-Feature Parkinson's Research | Python, Pandas, scikit-learn, XGBoost

- Derived a vocal instability index (**VIC**) on the UCI Oxford Parkinson's dataset (195 recordings, 32 subjects); alone it reaches **0.833 ROC-AUC** vs 0.734 for the raw 22 features. Inspired by, but not reusing, the fPI index of Gupta, Bhan and Nimsarkar.
- Compared five classifiers under subject-grouped 5-fold CV over 10 seeds; reported XGBoost against a majority baseline (accuracy 0.796 vs 0.756).

Employee Attrition Analysis | Python, Pandas, scikit-learn, XGBoost, SHAP

- Predicted attrition on the IBM HR dataset (1,470 employees, 16% attrition) using SMOTE on the training split; XGBoost reached 0.759 ROC-AUC on a 30% held-out test set.
- Ranked attrition drivers with **SHAP** (overtime first) and reported the model's low recall (0.30) as a limitation.

PUBLICATIONS

V-Surveillance: A Hybrid Deep Learning Framework for Real-Time Aerial Surveillance Using Drone Imagery — IEEE CICT (first author) — Dec 2025

An Enhanced Object-Oriented Programming-Based Web Page Linker — IEEE IATMSI (co-author) — Mar 2024

EDUCATION

Dr. Shyama Prasad Mukherjee International Institute of Information Technology, Naya Raipur

2022 – 2026

B.Tech in Data Science and Artificial Intelligence | CGPA 7.57 / 10 (80.7%, official conversion)

Completed July 2026

- Coursework: Deep Learning, Computer Vision, Optimization Methods in ML, NLP, Signal and System; Undergraduate Research Work I-II.

TECHNICAL SKILLS

Languages: Python, C, TypeScript, JavaScript, SQL, Dart, Bash

Machine Learning & CV: PyTorch, Torchvision, scikit-learn, XGBoost, SHAP, Pandas, YOLO12, SAHI, ESRGAN, DeepSORT

Backend & Systems: FastAPI, Docker, Linux, DuckDB, PostgreSQL, C11 atomics, multithreading, RS-422, WebSockets

Frontend & Mobile: React, Vite, Flutter, Zustand, Dexie.js (IndexedDB), TanStack Virtual